Supplementary material

Healthy-Data-Only, Gradient-Free On-Device Recalibration of an Integer-Only Vibration Fault Classifier on a 20 KB-RAM Microcontroller

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This document contains material that supports the paper but was moved out of the main text to meet the length limit of the journal. All numbers are produced by the released scripts from the result files of the five-seed runs; ± denotes the population standard deviation.

S1. Validation of the theory on synthetic data

To test Propositions 1–3 and the integer implementation under a known shift, three-class 16 kHz vibration signals were synthesised: the healthy class contains shaft harmonics and noise, the outer-race class adds periodic decaying impulses, and the inner-race class adds impulses whose amplitude is modulated at the shaft frequency. Five surrogate models with the architecture of Table 3 of the paper were built with random convolution weights and a ridge-regression output layer, and quantised to int8. The shift was applied in the first layer by multiplying its weights by γ and replacing its bias by $\gamma b + \delta$. The shifted weights are held as wider integers (beyond the int8 range for $\gamma > 1$), so this is an idealised check of the derivation and the code, not of int8 deployment. With integer γ and δ the shift is exactly affine; with continuous $\gamma \in [0.5, 2]$, weight rounding makes it only approximately affine. These results validate the theory and the code and say nothing about diagnostic performance on real bearings.

Table S1. Validation on synthetic data (5 surrogate models, mean ± std, %).

Setting	Method	Accuracy	Agreement with source model
No shift, no calibration (source model)	–	82.4 ± 3.5	100.0
Exact affine shift ($\gamma \in \{1, 2, 3, 4\}$)	M0	33.7 ± 0.7	34.7 ± 1.4
Exact affine shift, exact statistics	M3L1	82.4 ± 3.4	99.7 ± 0.3
Approximate affine shift ($\gamma \in [0.5, 2]$)	M0	41.6 ± 15.5	41.6 ± 15.5
Approximate affine shift, $N = 32$	M1	47.2 ± 3.0	52.6 ± 6.0
Approximate affine shift, $N = 32$	M3	79.2 ± 3.6	86.4 ± 3.7
No shift, exact statistics	M1	49.6 ± 2.2	54.4 ± 5.4
No shift, exact statistics	M3	82.4 ± 3.5	100.0 ± 0.0
No shift, $N = 32$	M1	46.1 ± 2.6	51.7 ± 5.8
No shift, $N = 32$	M3	79.5 ± 4.0	86.9 ± 3.4
$\gamma = 6$ (beyond the clip limit of 4)	M0 / M3L1	33.3 ± 0.0 / 52.4 ± 1.6	33.9 ± 1.3 / 53.9 ± 2.1

Under an exact affine shift, recalibration restores the agreement with the source model from 34.7% to 99.7%, the remainder being due to fixed-point rounding (Proposition 1). Without a shift and with exact statistics, the healthy anchor leaves the model unchanged, whereas the all-class anchor lowers accuracy from 82.4% to 49.6% (Proposition 2). With 32 windows per layer, estimation noise lowers the agreement of M3 to about 87%. When fault windows are mixed into the calibration data, the error of the estimated mean of an intermediate layer grows linearly with ε ($R^2 = 0.998$, Figure S1a), as predicted by Proposition 3; this checks the implementation rather than providing independent evidence. Accuracy hardly changes for $\varepsilon \leq 0.1$ and falls to 57.5% at $\varepsilon = 0.5$ (Figure S1b). A shift beyond the clipping range is only partly recovered.

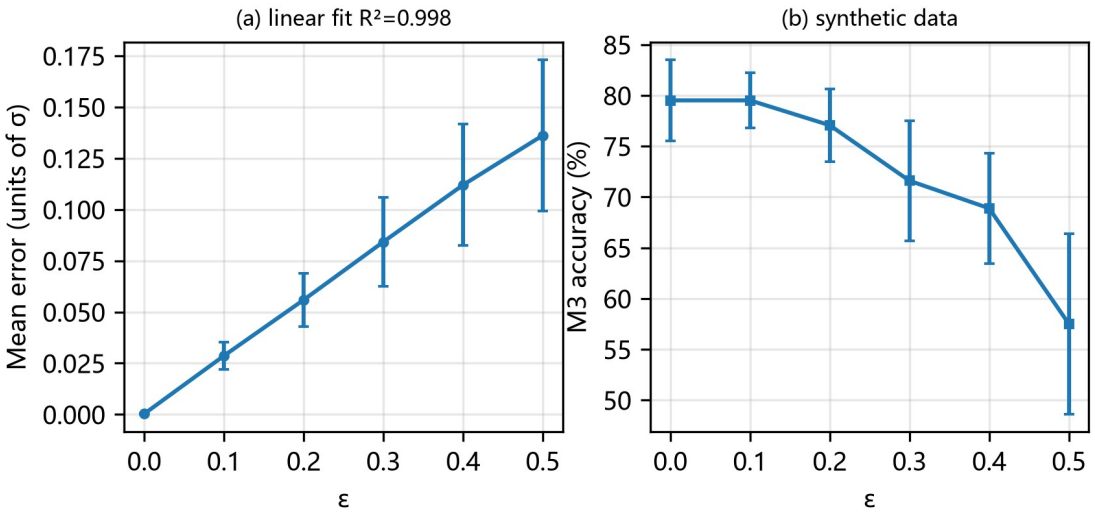


Figure S1. Proposition 3 on synthetic data: (a) fraction ε of fault windows versus the error of an intermediate-layer mean (in units of the healthy standard deviation); (b) ε versus M3 accuracy. Mean ± std over five surrogate models.

S2. Sensitivity of α-BN to α

Table S2. Accuracy of α-BN for different α (%, mean over 5 seeds); α is selected by the highest development mean.

Target	α = 0.05	α = 0.1	α = 0.2	α = 0.3	α = 0.5	M3L1
C1 (development)	27.7	31.2	31.4	29.7	27.5	11.9
C2-1HP (development)	84.8	74.0	54.1	44.8	41.3	88.4
P1 (development)	68.9	84.8	86.7	78.0	70.2	82.3
Development mean	60.5	63.4	57.4	50.8	46.3	60.9
C2-2HP (held out)	80.0	71.2	52.2	42.8	39.4	84.5
C2-3HP (held out)	76.4	68.7	53.1	43.8	40.4	83.3
P2 (held out)	99.2	96.4	85.2	77.3	73.5	99.8
P3 (held out)	83.2	84.0	74.1	64.9	60.0	80.0
Held-out mean	84.7	80.1	66.1	57.2	53.3	86.9

S3. Macro-F1

Table S3. Macro-F1 on each target (%, mean ± std over 5 seeds).

Target	M0	M1	M3	M3L1	B2*	M4
C1	13.4 ± 3.2	21.0 ± 3.5	4.9 ± 3.4	6.0 ± 2.6	24.7 ± 4.5	93.0 ± 1.5
C2-1HP	89.1 ± 3.6	30.6 ± 12.3	71.2 ± 12.1	86.6 ± 5.1	69.6 ± 7.9	98.6 ± 0.4
C2-2HP	86.1 ± 5.3	28.5 ± 12.0	69.2 ± 17.0	81.1 ± 7.7	65.4 ± 7.0	98.7 ± 0.5
C2-3HP	80.4 ± 5.4	29.4 ± 12.6	55.4 ± 16.5	80.7 ± 7.9	62.1 ± 4.9	98.8 ± 0.4
P1	39.6 ± 11.0	66.7 ± 3.3	77.8 ± 6.1	82.0 ± 4.7	84.3 ± 4.2	96.7 ± 0.3
P2	99.8 ± 0.1	69.6 ± 2.6	99.8 ± 0.1	99.8 ± 0.1	96.4 ± 1.0	99.9 ± 0.1
P3	73.2 ± 5.6	57.6 ± 3.2	75.8 ± 6.6	78.9 ± 6.4	83.7 ± 2.7	99.4 ± 0.2
Mean	68.8	43.3	64.9	73.6	69.5	97.9

S4. Additional figures

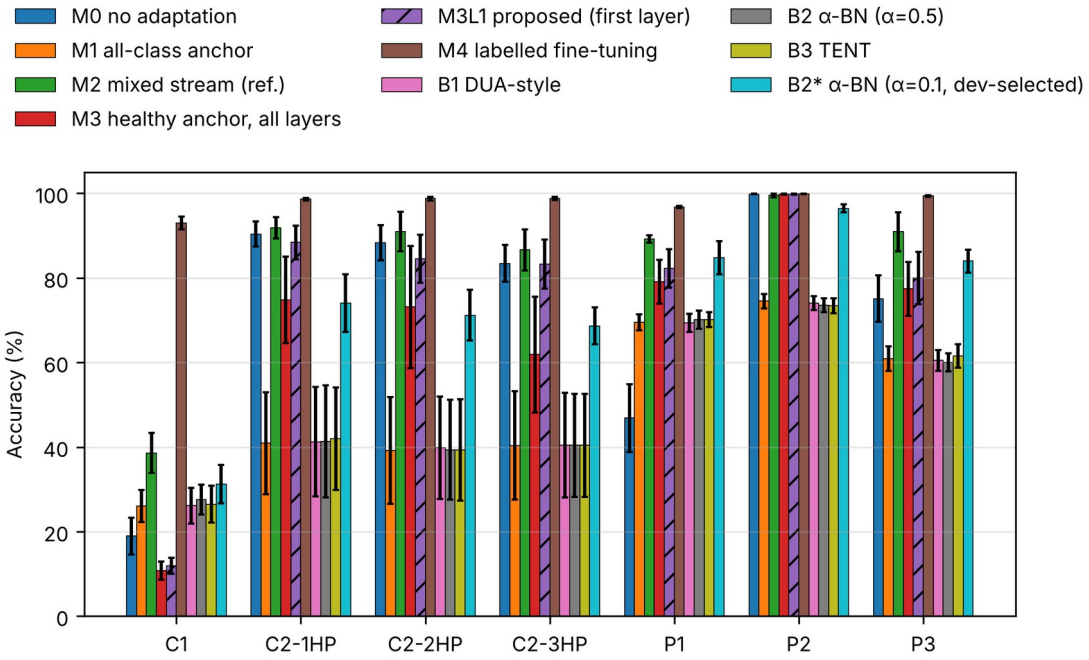


Figure S2. Accuracy of all methods on the seven target domains (mean ± std over 5 seeds); the data are those of Table 6 of the paper.

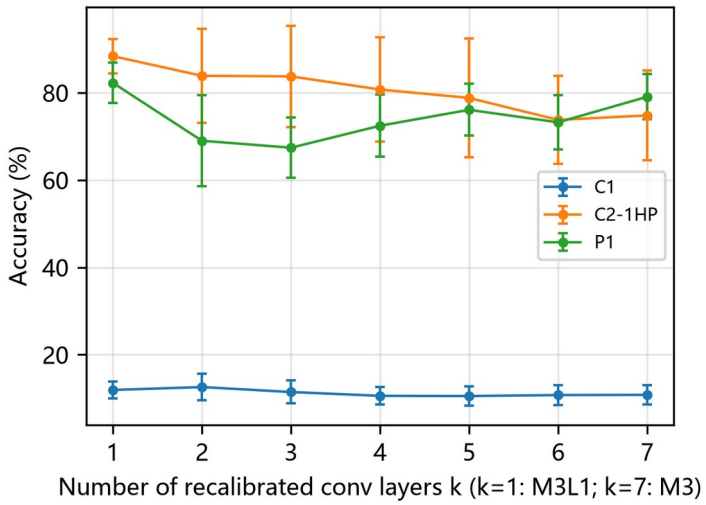


Figure S3. Accuracy versus the number of recalibrated layers k on the development targets (mean ± std over 5 seeds); the data are those of Table 8 of the paper.

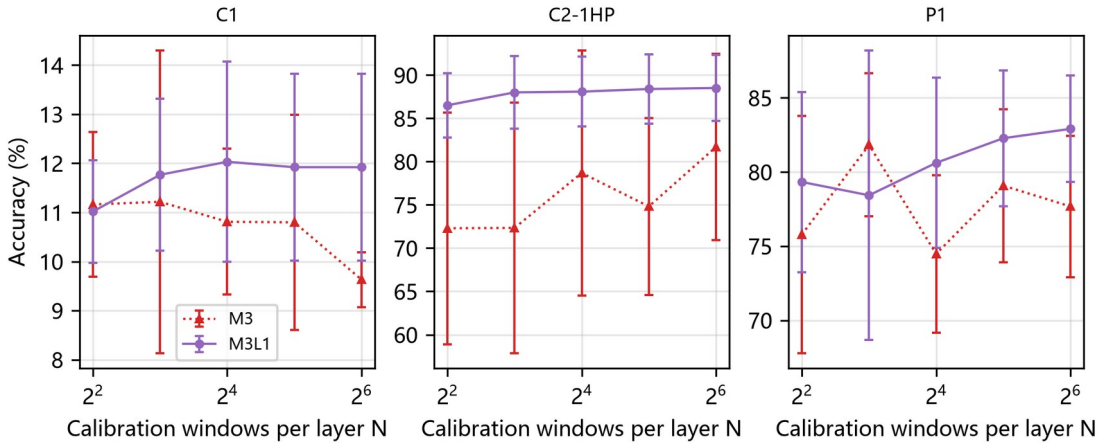


Figure S4. Accuracy versus the number of calibration windows per layer N on the development targets (mean ± std over 5 seeds).

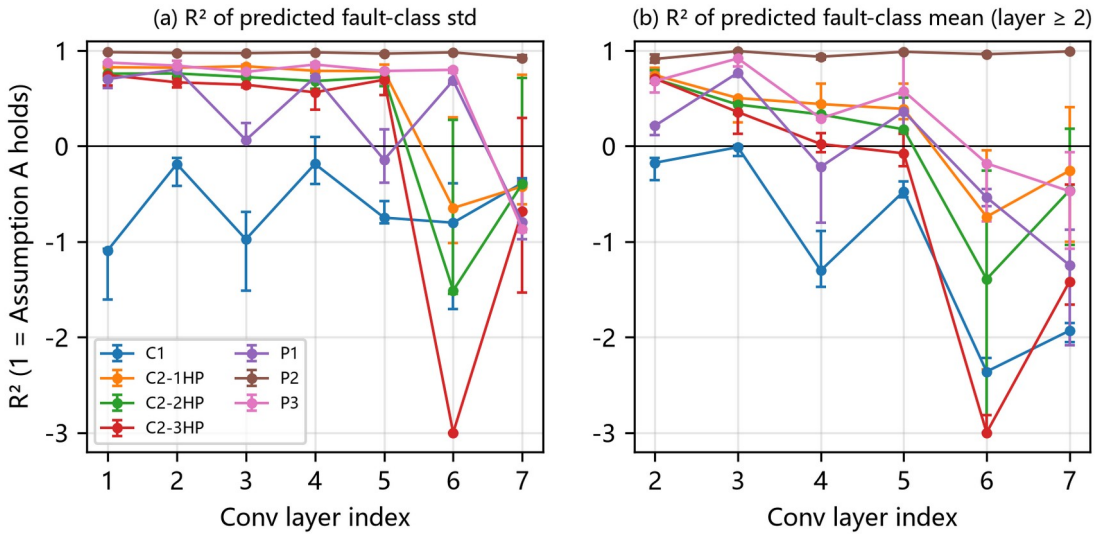


Figure S5. Test of Assumption A: R^2 of the predicted fault-class standard deviation (a) and mean (b, from layer 2) per layer (median of 5 seeds; bars show the interquartile range; values below −3 are drawn at the lower bound).

S5. M1L1 control on all targets

M1L1 recalibrates only the first layer, like M3L1, but aligns to the all-class anchor. It uses the same source models, the same 32 calibration windows per target and seed and the same integer update as M3L1; the recomputed M3L1 values are identical to those of Table 6 of the paper.

Table S4. M1L1 versus M3L1 on all targets (accuracy, %, mean ± std over 5 seeds). D: development; H: held-out.

Target	M0	M1L1	M3L1
C1 (D)	18.9 ± 4.3	21.3 ± 3.2	11.9 ± 1.9
C2-1HP (D)	90.4 ± 3.0	46.4 ± 15.6	88.4 ± 4.0
P1 (D)	46.8 ± 8.0	72.4 ± 4.3	82.3 ± 4.6
C2-2HP (H)	88.3 ± 4.1	44.0 ± 13.0	84.5 ± 5.7
C2-3HP (H)	83.5 ± 4.3	47.4 ± 13.4	83.3 ± 5.8
P2 (H)	99.8 ± 0.1	73.3 ± 2.6	99.8 ± 0.1
P3 (H)	75.1 ± 5.5	66.2 ± 4.7	80.0 ± 6.2
Mean, all 7	71.8	53.0	75.7
Mean, held-out 4	86.7	57.7	86.9

Over all 35 (target, seed) pairs M3L1 is 22.8 points better than M1L1 (30 wins, 5 losses, all on C1; two-sided Wilcoxon $p = 4.9 \times 10^{-6}$ with the normal approximation used for Table 7 of the paper, and 4.9×10^{-5} after Bonferroni correction over ten comparisons); over the 20 held-out pairs it is 29.2 points better (20 wins). With the four held-out targets as units M3L1 wins on all four (exact Wilcoxon $p = 0.125$, the smallest value possible with four units). In the first layer the means of the two anchors almost coincide (Table 9 of the paper), but the ratio of their standard deviations per channel ranges from 0.30 to 2.34 over all sources and seeds (median 0.87–1.19 per source), so M1L1 rescales the healthy data with a wrong gain. Without a shift (protocol of Table 5 of the paper), M1L1 gives 56.80 ± 19.41% (C1 source), 59.95 ± 21.62% (C2 source) and 78.55 ± 3.52% (PU source), whereas M3L1 gives 99.85 ± 0.09%, 99.54 ± 0.25% and 99.87 ± 0.12%.

S6. Check on a third dataset (JNU)

Data and protocol. The Jiangnan University (JNU) bearing dataset (vertical acceleration sampled at 50 kHz; files n600_3_2, n800_3_2, n1000_3_2, ib600_2, ib800_2, ib1000_2, ob600_2, ob800_2, ob1000_2, tb600_2, tb800_2 and tb1000_2 (.csv) from https://github.com/ClarkGableWang/JNU-Bearing-Dataset; the dataset is usually cited through reference [26] of the paper, which describes a Jiangnan University test rig with the same fault types and 50 kHz sampling but reports experiments with speed varying within about 400–800 r/min; the constant-speed files used here, including those at 1000 r/min, are taken from the repository above and are not described in [26]) contains healthy, inner-race, outer-race and roller-fault records at 600, 800 and 1000 r/min. The first 500 500 samples (10 s) of each record are used, so that all classes have equal length. The 1000 r/min records form the source domain, split in time into 60% training, 20% validation and 20% test. The 800 r/min (J1) and 600 r/min (J2) records are two held-out targets: the first 30% of each record is the calibration segment, of which only the healthy windows are used, and the remaining 70% is the test segment. Each

segment is decimated by 3 (anti-aliasing FIR filter, 16.7 kHz) on its own, so that no raw sample is shared between segments, and the int16 scale is set on the source training segment. Windows have 1024 samples and a stride of 256 in the source training segment and 512 elsewhere (1552, 256 and 256 source windows; per target 384 calibration windows, 96 of them healthy, and 908 test windows). The network, training, quantisation, methods and parameters ($N = 32$, $\alpha = 0.5$ for B2, $\alpha = 0.1$ for B2*) are those of the paper, without any change. These settings, the quantities to be reported and the decision rule (M3L1 above M0 on both targets) were written down before any result was obtained.

Table S5. JNU check (accuracy, %, mean ± std over 5 seeds; macro-F1 as mean). "No shift": calibration on healthy source validation windows, test on the source test segment (256 windows, one window = 0.39 points). †: uses data that are normally unavailable on site. –: not evaluated.

Method	No shift (source test)	J1 (800 r/min)	J2 (600 r/min)	Macro-F1 J1 / J2
M0 (no adaptation)	97.2 ± 0.9	88.9 ± 2.2	51.2 ± 5.2	88.7 / 48.1
M1 (all-class anchor, 7 layers)	64.8 ± 7.9	65.1 ± 8.3	44.7 ± 8.5	62.0 / 40.2
M1L1 (all-class anchor, layer 1)	79.6 ± 5.9	73.7 ± 4.5	45.4 ± 3.1	70.5 / 38.6
M2 (class-mixed data)†	–	91.4 ± 1.3	68.9 ± 2.2	91.4 / 68.5
M3 (healthy anchor, 7 layers)	93.0 ± 1.5	82.4 ± 4.8	49.3 ± 7.0	81.5 / 42.6
M3L1 (proposed)	97.2 ± 1.1	91.6 ± 0.9	49.1 ± 5.2	91.6 / 46.9
M4 (labelled fine-tuning)†	–	95.4 ± 0.5	80.1 ± 2.6	95.4 / 80.0
B1 (DUA, simplified)	–	64.6 ± 7.7	45.9 ± 9.4	61.2 / 41.1
B2 (α-BN, α = 0.5)	–	66.9 ± 6.1	48.6 ± 9.2	63.3 / 43.6
B2* (α-BN, α = 0.1)	86.7 ± 4.2	80.8 ± 3.3	58.1 ± 2.3	79.8 / 54.8
B3 (TENT)	–	63.7 ± 9.0	43.4 ± 7.2	60.6 / 38.2

Result against the decision rule. M3L1 is above M0 on J1 (+2.7 points; 4 wins, 1 tie, 0 losses) but below M0 on J2 (−2.1 points; 1 win, 4 losses, largest loss 9.4 points for one seed). The predefined rule is therefore not met, and the method is not consistently supported on this dataset. Over the 10 (target, seed) pairs, M3L1 is 0.3 points above M0 (exact two-sided Wilcoxon test on the nine non-zero differences, $p = 0.91$).

Comparison with the other methods (10 pairs, exact Wilcoxon tests, ties discarded, not corrected for multiple comparisons). M3L1 is better than M1 by 15.4 points (8 wins, 2 losses, $p = 0.020$), than M1L1 by 10.8 points (9 wins, 1 loss, $p = 0.0059$), than B1, B2 and B3 by

15.1, 12.6 and 16.8 points ($p = 0.027$, 0.049 and 0.014) and than M3 by 4.5 points ($p = 0.16$). Against B2* it is 0.9 points better on average (5 wins, 5 losses, $p = 0.92$): better by 10.8 points on J1 but worse by 9.1 points on J2. On J1, M3L1 is 18–28 points above M1, M1L1, B1, B2 and B3 on average and better than each of them in all five seeds. With $\alpha = 0.05$, 0.1, 0.2 and 0.3, B2 reaches 85.0, 80.8, 74.6 and 70.5% on J1 and 57.4, 58.1, 55.6 and 52.3% on J2.

No shift. M3L1 leaves the mean accuracy unchanged (97.2% for both; per seed −0.78 to +1.56 points, i.e. −2 to +4 test windows), whereas M1 lowers it to 64.8% (52.0–75.0% per seed) and M1L1 to 79.6%, as Proposition 2 predicts; the per-channel ratio of the first-layer standard deviations of the two anchors is 0.58–0.87 (median 0.68). B2* lowers it to 86.7%.

Diagnostics. With the procedure of Section 5.4 of the paper (seed 0), the first-layer R^2 of the predicted fault-class standard deviation is 0.70 on J1 but only 0.06 on J2; the corresponding seed-0 values for the targets of the paper are 0.70 (P1), 0.88 (P3), 0.80–0.88 (CWRU load targets) and −1.72 (C1). The 600 r/min change is thus far from a per-channel gain change in the first layer, and M3L1 did not correct it. The source model classifies only 11% of the healthy calibration windows of J2 as healthy (76% on J1); after M3L1 the fraction rises to 62% (94% on J1), but the overall accuracy does not improve. In the single-seed ablation on J1, $k = 1$ is again the best number of recalibrated layers (92.1% against 86.2–91.0% for $k = 2$–7), and M3L1 stays at 90.7–92.4% for fault-window fractions up to $\varepsilon = 0.5$, whereas M3 falls to 26.7%.

On-board check. The seed-0 JNU model was also exported, compiled and run on the STM32F103C8T6 with the procedure of Section 5.5 of the paper (35 164 B Flash, 16 104 B static RAM, 106.21 ms per inference). On the 256 source test windows the logits of the board were bit-identical to the Python reference. On J1 and J2 the board calibrated the first layer itself; its parameters (16/16 channels on each target) and the logits of 300 test windows before and after calibration were bit-identical to the reference. Board accuracy changed from 94.7% to

92.7% on J1 and from 47.3% to 43.0% on J2 (one model, differently drawn anchors; Table 11 of the paper).

Interpretation. The check reproduces the analysis of the paper on a third test rig: conventional recalibration with healthy-only data degrades even without a shift, and the healthy anchor removes this degradation. It also reproduces a gain over M0 for a moderate speed change, but not for the larger speed change, where the first-layer change violates Assumption A. This agrees with the scope stated in the paper: the method is a correction for shifts that are approximately per-channel gain changes in the first layer, not a general solution to domain shift.

S7. Paired differences per target

Table 7 of the paper uses two-sided Wilcoxon signed-rank tests on (target, seed) pairs with the normal approximation and zero differences split between positive and negative ranks (SciPy 1.17.1, zero_method = ‘zsplit’, method = ‘approx’), with Holm’s correction within each set of comparisons; these pairs are not fully independent. Table S6 therefore gives the paired difference of every target separately.

Table S6. Paired accuracy differences per target (percentage points; mean ± std over 5 seeds of the per-seed difference, with wins / ties / losses of M3L1). D: development; H: held-out; J1 and J2: JNU check (Section S6). Differences are computed from unrounded accuracies, so they can differ by 0.1 from differences of the rounded values in Table 6 of the paper (for example, +35.4 on P1).

Target	M3L1 − M0	M3L1 − M1	M3L1 − M1L1	M3L1 − B2*
C1 (D)	−7.0 ± 3.9 (0/0/5)	−14.1 ± 5.4 (0/0/5)	−9.3 ± 3.7 (0/0/5)	−19.3 ± 3.6 (0/0/5)
C2-1HP (D)	−2.0 ± 3.5 (1/0/4)	+47.5 ± 13.1 (5/0/0)	+42.0 ± 18.4 (5/0/0)	+14.4 ± 6.8 (5/0/0)
P1 (D)	+35.4 ± 5.3 (5/0/0)	+12.7 ± 5.8 (5/0/0)	+9.9 ± 5.6 (5/0/0)	−2.5 ± 3.6 (1/0/4)
C2-2HP (H)	−3.8 ± 5.6 (1/0/4)	+45.3 ± 14.9 (5/0/0)	+40.6 ± 17.5 (5/0/0)	+13.3 ± 7.6 (5/0/0)
C2-3HP (H)	−0.2 ± 5.5 (3/0/2)	+42.9 ± 14.9 (5/0/0)	+35.9 ± 18.1 (5/0/0)	+14.6 ± 6.3 (5/0/0)
P2 (H)	0.0 ± 0.0 (1/0/4)	+25.3 ± 1.6 (5/0/0)	+26.5 ± 2.6 (5/0/0)	+3.4 ± 0.9 (5/0/0)
P3 (H)	+4.9 ± 1.5 (5/0/0)	+19.1 ± 5.0 (5/0/0)	+13.8 ± 6.9 (5/0/0)	−4.0 ± 4.5 (1/0/4)
J1 (JNU)	+2.7 ± 1.8 (4/1/0)	+26.5 ± 7.7 (5/0/0)	+17.9 ± 4.0 (5/0/0)	+10.8 ± 2.6 (5/0/0)
J2 (JNU)	−2.1 ± 7.0 (1/0/4)	+4.4 ± 11.7 (3/0/2)	+3.7 ± 4.9 (4/0/1)	−9.1 ± 5.8 (0/0/5)

With the seven main targets as units, M3L1 is better than M1 and M1L1 on six of seven targets (two-sided sign test $p = 0.125$), better than M0 on two ($p = 0.45$) and better than B2* on four ($p = 1.0$). With so few units such tests have little power; they show that the significance of the pair-level tests depends on treating seeds as replicates and that the differences are consistent within targets rather than across all targets.

S8. Gain limits and confusion matrices

Gain limits. The first-layer gain a of M3L1 is limited to [1/4, 4] (Section 3.3 of the paper). Over five seeds, the seven main targets (560 channel estimates), J1 and J2 (160) and the no-shift checks of the four source models (320), the limits were never reached and no estimated standard deviation was zero; a ranged from 0.55 to 1.69 on the main targets, from 0.63 to 1.94 on J1 and J2 and from 0.91 to 1.08 without a shift. The limits therefore did not affect any reported result. The script clip_and_confusion.py of the code repository reproduces these numbers from the saved source models; it also recomputes the accuracies of M0, M1 and M3L1, which agree exactly with Table 6 of the paper and Table S5 (135 of 135 target–seed–method values).

Confusion matrices. Table S7 gives the false-alarm rate (healthy windows classified as a fault) and the missed-detection rate (fault windows classified as healthy) of M0, M1 and M3L1, and Figure S6 the confusion matrices of four targets, all summed over the five seeds.

Table S7. Accuracy, false alarms and missed detections (%, summed over 5 seeds; values for M0 / M1 / M3L1). D: development; H: held-out.

Target	Accuracy	False alarms	Missed detections
C1 (D)	18.9 / 26.0 / 11.9	0.2 / 92.4 / 0.0	57.5 / 0.2 / 73.5
C2-1HP (D)	90.4 / 40.9 / 88.4	29.1 / 99.0 / 0.0	0.0 / 0.0 / 0.0
P1 (D)	46.8 / 69.5 / 82.3	0.0 / 65.5 / 0.0	78.4 / 0.0 / 18.7
C2-2HP (H)	88.3 / 39.2 / 84.5	33.8 / 99.4 / 0.0	0.0 / 0.0 / 0.0
C2-3HP (H)	83.5 / 40.4 / 83.3	40.7 / 99.6 / 0.0	0.0 / 0.0 / 0.1
P2 (H)	99.8 / 74.5 / 99.8	0.1 / 72.6 / 0.2	0.0 / 0.0 / 0.0
P3 (H)	75.1 / 60.9 / 80.0	0.0 / 70.2 / 0.0	21.3 / 0.0 / 14.6
J1 (JNU)	88.9 / 65.1 / 91.6	21.9 / 73.4 / 7.0	1.8 / 0.0 / 4.4
J2 (JNU)	51.2 / 44.7 / 49.1	91.3 / 80.1 / 45.7	1.4 / 4.8 / 10.7

- M1 classifies 65–100% of the healthy windows of the main targets as faults, as Proposition 2 predicts.
- On the CWRU load targets M0 classifies 29–41% of the healthy windows as faults. M3L1 removes these false alarms with at most 0.1% missed detections, but confuses more fault types (on C2-3HP the recall of IR007 falls from 99% to 68% and that of IR014 from 67% to 55%), so its accuracy is slightly below that of M0.
- On P1 and P3 M3L1 keeps the false alarms below 0.1% and raises the recall of both fault classes (P1: outer race 26% → 72%, inner race 15% → 74%; P3: 79% → 83% and 46% → 57%); its gain there is therefore not only a better separation of healthy and faulty windows.
- On C1 M3L1 classifies 73% of the fault windows as healthy (M0: 58%), the more dangerous failure mode in practice.
- On J2 M3L1 halves the false alarms of M0 (91% → 46%) but raises the missed detections from 1.4% to 10.7% and lowers the outer-race recall from 50% to 17%.

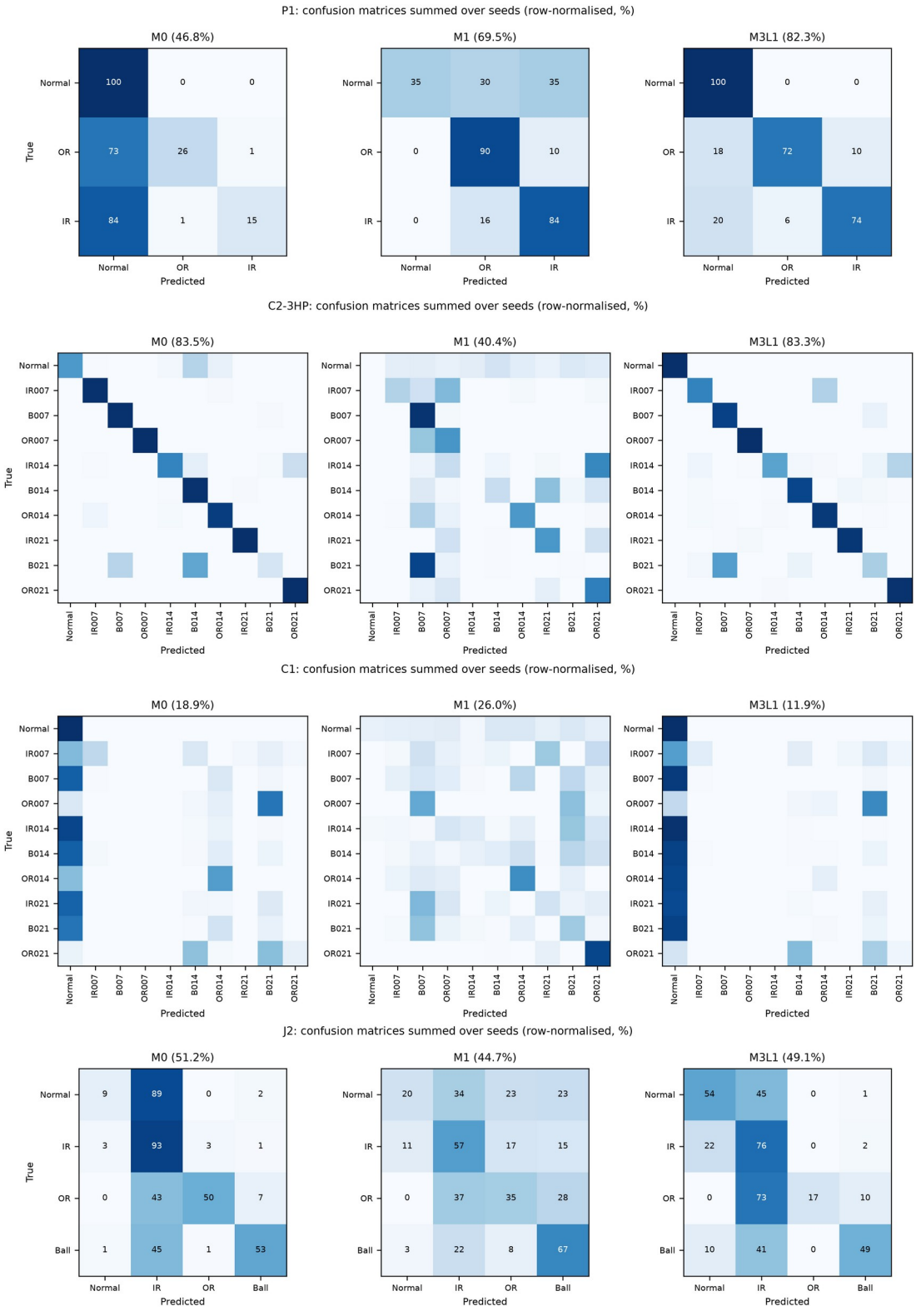


Figure S6. Confusion matrices of M0, M1 and M3L1 on P1, C2-3HP, C1 and J2 (summed over five seeds; row-normalised, %; accuracy in parentheses).